

Week 11: Fitting, Optimization, and Strategic Games

Linear Regression, Taylor Expansion, Nash Equilibrium, and Pareto Thinking

Instructor / Mentor

Kiki Research Training Program

Week 11

Contents

1. Class Mission
2. Fitting as Optimization
3. Taylor Expansion
4. Neural Networks as Optimization
5. Game Theory and Demand Response
6. Pareto Thinking
7. Class Lab and Wrap-Up

Class Mission

Week 11 Mission

Core idea

Optimization is not only about dispatching generators. It is also the hidden engine behind **fitting models** and many **strategic decisions** in smart grids.

1. Continue optimization by asking: “Which parameters best fit the data?”
2. Introduce linear regression as the simplest supervised learning model.
3. Use Taylor expansion as a local approximation: a small function fitted near one point.
4. Preview neural networks as huge parametric functions trained by optimization.
5. Introduce game theory, Nash equilibrium, and Pareto thinking through demand response.

Today's promise

By the end, students should be able to say: “Fitting means choosing parameters that make model predictions close to data.”

120-Minute Flow

1. **0–10 min:** recall Week 10 optimization language.
2. **10–35 min:** fitting with predefined parametric functions.
3. **35–55 min:** linear regression and least-squares loss.
4. **55–70 min:** Taylor expansion as local fitting.
5. **70–80 min:** neural networks as large optimization problems.
6. **80–105 min:** demand response game and Nash equilibrium.
7. **105–115 min:** Pareto optimality and social tradeoffs.
8. **115–120 min:** exit ticket and homework briefing.

Why Optimization Is Not Finished

Last week

We optimized **physical decisions**:

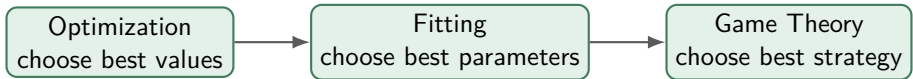
- generator outputs,
- total cost,
- demand-balance constraints,
- capacity limits.

This week

We optimize **model parameters** and **human strategies**:

- slope and intercept of a fitted line,
- weights of a neural network,
- response choices under incentives,
- tradeoffs among different goals.

The Bridge for Today



One repeated question

What counts as “best”? The answer depends on the objective: lowest cost, smallest prediction error, highest payoff, or best tradeoff.

Fitting as Optimization

What Does “Fitting” Mean?

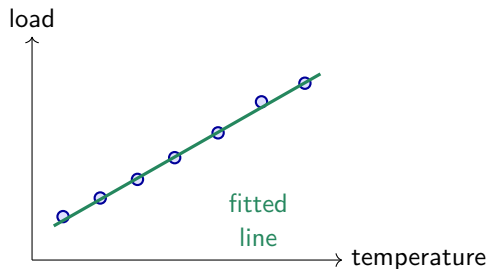
Fitting idea

We choose a model shape first, then adjust its parameters so the model matches the data as well as possible.

data points $\longrightarrow$ best parameter values

Everyday version

A student draws a line through dots on a graph. Linear regression asks the computer to choose the line more carefully.



Predefined Parametric Function

Important phrase

A **predefined parametric function** is a function family with adjustable numbers called parameters.

Model shape	Function	Parameters
Constant	$\hat{y} = c$	c
Line	$\hat{y} = wx + b$	w, b
Quadratic curve	$\hat{y} = ax^2 + bx + c$	a, b, c
Neural network	$\hat{y} = f(x; \theta)$	many weights θ

Research habit

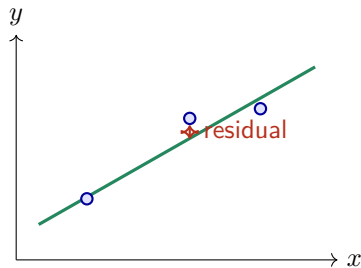
The model shape is an assumption. A fitted answer can look precise even when the assumption is wrong.

Residuals: The Model's Mistakes

For each data point:

$$\text{residual} = y_i - \hat{y}_i$$

- y_i is the real measured value.
- $\hat{y}_i$ is the model prediction.
- Small residuals mean the model is close to the data.



Loss Function: Turn Many Mistakes Into One Score

Optimization needs one score

To optimize a fitted model, we turn all residuals into a single number called a **loss function**.

$$\text{MSE}(w, b) = \frac{1}{n} \sum_{i=1}^n (y_i - (wx_i + b))^2$$

Why square the errors?

- Positive and negative errors do not cancel.
- Large mistakes are punished more.
- The math becomes smooth and optimizer-friendly.

Optimization form

$$\min_{w, b} \frac{1}{n} \sum_{i=1}^n (y_i - (wx_i + b))^2$$

Linear Regression: The Simplest Fitting Model

Model

$$\hat{y} = wx + b$$

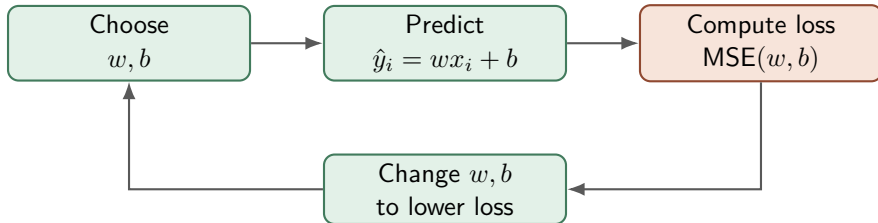
The slope w says how much $\hat{y}$ changes when x increases by 1. The intercept b is the prediction when $x = 0$.

Smart-grid example

If x is outside temperature and y is electricity load, linear regression asks:

Which slope and intercept make the load predictions closest to measured load?

Fitting a Line Is an Optimization Problem



Same template as dispatch

Dispatch chose generator outputs to minimize cost. Linear regression chooses model parameters to minimize prediction error.

Manual Example: Which Line Fits Better?

Hour x	Load y	Units
1	52	MW
2	56	MW
3	61	MW
4	65	MW

Compare two candidate models:

$$\hat{y} = 4x + 48$$

$$\hat{y} = 2x + 52$$

Question

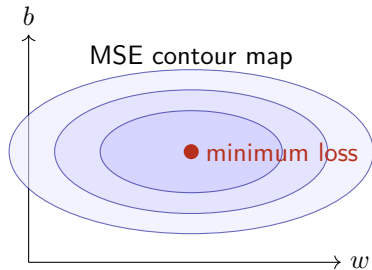
Which one has the smaller mean squared error?

The Loss Surface Picture

Parameter space

In linear regression with one input, every point (w, b) represents one possible line.

- The loss surface assigns an MSE to every (w, b) .
- Training means moving toward lower loss.
- The best-fit line is the lowest point on this surface.



Python Mini Lab: Loss for a Line

```
1 import numpy as np
2
3 x = np.array([1, 2, 3, 4], dtype=float)
4 y = np.array([52, 56, 61, 65], dtype=float)
5
6 def mse(w, b):
7     y_hat = w * x + b
8     error = y - y_hat
9     return np.mean(error ** 2)
10
11 print(mse(4, 48))
12 print(mse(2, 52))
```

Discussion

The code does not “know” electricity. We translate electricity data into numbers, then define what a good fit means.

Taylor Expansion

Taylor Expansion: A Local Fit

Simple intuition

Taylor expansion fits a complicated function near one point using a simpler function.

$$f(x) \approx f(a) + f'(a)(x - a)$$

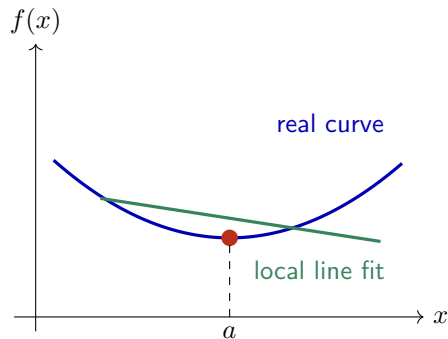
First-order approximation

Use the value and slope at a to create a nearby line.

Key warning

Taylor expansion is local. It can be accurate near a but bad far away.

Taylor Picture: Curve and Tangent Line



Second-Order Taylor: Add Curvature

Better local shape

A second-order approximation also uses curvature:

$$f(x) \approx f(a) + f'(a)(x - a) + \frac{1}{2}f''(a)(x - a)^2$$

- $f(a)$ says where the function is.
- $f'(a)$ says which direction it is moving.
- $f''(a)$ says how the slope is changing.

Optimization connection

Many optimizers use local slope and curvature information to decide how to update parameters.

Why Taylor Expansion Helps Optimization

Hard problem

A real loss function may be complicated:

$$\text{Loss}(\theta)$$

Instead of understanding the whole shape at once, we approximate what happens near the current parameters.

Local question

- If I change θ a tiny bit, does loss go up or down?
- Which direction seems best nearby?
- How big should the next step be?

Taylor Expansion vs. Linear Regression

Idea	What it fits	Where it fits
Linear regression	a line to data points	across a dataset
Taylor expansion	a simple function to another function	near one point
Neural network	a flexible function to data points	across a dataset

Common theme

All three are about replacing something complicated with a function that is useful, adjustable, or easier to compute.

Neural Networks as Optimization

Neural Networks Are Also Fitted Functions

Same idea, bigger model

A neural network is a parametric function:

$$\hat{y} = f(x; \theta)$$

where θ represents many weights and biases.

Linear regression

$$\hat{y} = wx + b$$

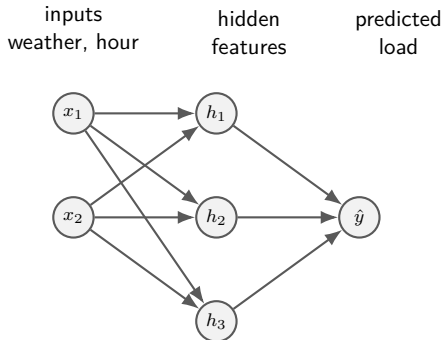
Maybe 2 parameters.

Neural network

$$\hat{y} = f(x; \theta)$$

Could have thousands, millions, or billions of parameters.

Tiny Neural Network Diagram



Training

Training means changing all weights so prediction error gets smaller.

Training Loop: Same Optimization Template

1. Choose initial parameters θ .
2. Use the model to make predictions $\hat{y} = f(x; \theta)$.
3. Compute loss, such as MSE.
4. Estimate which parameter changes reduce loss.
5. Update parameters and repeat many times.

Important message for students

Neural networks are powerful, but they are not magic. They are fitted functions with many adjustable parameters.

Why Neural-Network Optimization Is Harder

Linear regression

- Few parameters.
- Smooth bowl-shaped loss for ordinary least squares.
- Easy to visualize.

Neural networks

- Many parameters.
- Complicated loss landscape.
- Training choices matter: learning rate, data, architecture.

Research mindset

When a model works, ask: Was it the data, the model shape, the loss function, or the optimization process?

Game Theory and Demand Response

From Optimization to Game Theory

Optimization assumption

One planner chooses the best decision for the whole system.

Game theory assumption

Multiple decision-makers choose their own actions, and each person's result depends on what others do.

Smart-grid translation

In demand response, the utility offers incentives, but households decide whether to reduce consumption based on comfort, payment, and what they expect others to do.

Demand Response Story

Scenario

At 5:00 PM, the grid is stressed. The utility sends a message:

"Reduce AC for one hour and receive a reward."

Each household asks

- How uncomfortable will I be?
- How much money do I get?
- If everyone else reduces, do I still need to?
- If nobody else reduces, is my action useful?

Payoff Matrix

Payoff

A payoff is a score for one player's outcome. Higher payoff means that player likes the outcome more.

		Household B	
		Reduce	Do not reduce
Household A	Reduce	(3, 3)	(1, 4)
	Do not reduce	(4, 1)	(0, 0)

Reading a cell

In (1, 4), Household A gets payoff 1 and Household B gets payoff 4.

Best Response

Definition

A player's **best response** is the action that gives the highest payoff after the other player's action is fixed.

If B reduces

A compares:

- A reduces: payoff 3.
- A does not reduce: payoff 4.

A's best response is not reducing.

If B does not reduce

A compares:

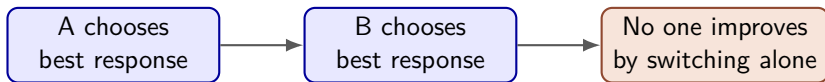
- A reduces: payoff 1.
- A does not reduce: payoff 0.

A's best response is reducing.

Nash Equilibrium

Definition

A **Nash equilibrium** is a stable strategy profile where no player wants to change alone.



Not the same as best for everyone

Nash equilibrium means individually stable. It does not always mean socially optimal or fair.

Find the Nash Equilibria in the Demand Game

		Household B	
		Reduce	Do not reduce
Household A	Reduce	(3, 3)	(1, 4)
	Do not reduce	(4, 1)	(0, 0)

1. Circle A's best response in each column.
2. Circle B's best response in each row.
3. A Nash equilibrium is a cell where both players are best-responding.

Interpretation

Two equilibria can exist: one household reduces while the other enjoys comfort. This is stable, but it may feel unfair.

Grid Design Question

Research question

How should the utility design incentives so that individual best responses also support the grid?

Possible tools

- rewards for reducing load,
- penalties during extreme peaks,
- randomized offers,
- community battery sharing,
- automated smart thermostats.

Design goal

Make the selfish choice close to the system-friendly choice.

Python Mini Lab: Pure Nash Equilibrium

```
1 import numpy as np
2
3 # rows: A actions, columns: B actions
4 A = np.array([[3, 1],
5               [4, 0]], dtype=float)
6 B = np.array([[3, 4],
7               [1, 0]], dtype=float)
8
9 for i in range(2):
10     for j in range(2):
11         a_best = A[i, j] == np.max(A[:, j]) # fix B's action
12         b_best = B[i, j] == np.max(B[i, :]) # fix A's action
13         if a_best and b_best:
14             print("Nash equilibrium at row", i, "column", j)
```

Coding idea

The code checks the definition directly: no player can improve by switching alone.

Pareto Thinking

Pareto Optimality: The Tradeoff Concept

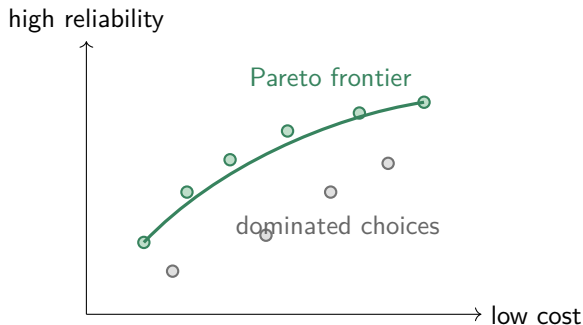
Definition

An outcome is **Pareto optimal** if we cannot make one person or objective better without making another worse.

Smart-grid examples

- Lower electricity cost vs. higher reliability.
- More renewable energy vs. more balancing cost.
- Customer comfort vs. peak-load reduction.
- Fast charging vs. transformer stress.

Pareto Frontier Picture



Concept only

Pareto thinking does not automatically choose one answer. It shows the tradeoff boundary where improvement in one direction has a cost.

Nash vs. Pareto

Concept	Main question	Smart-grid use
Nash equilibrium	Is anyone tempted to switch alone?	demand response strategies
Pareto optimality	Can we improve one goal without hurting another?	tradeoff design
Optimization	Which decision minimizes or maximizes an objective?	dispatch and fitting

Big lesson

A good grid decision may need all three: optimize the system, predict behavior, and understand tradeoffs.

Class Lab and Wrap-Up

Class Lab Part A: Fit a Load Line

1. Use four or five small data points: hour vs. electricity load.
2. Define a line $\hat{y} = wx + b$.
3. Write an MSE function.
4. Try multiple values of w and b .
5. Choose the smallest MSE and explain why that line fits best.

Manual thinking first

Students should predict which slopes are reasonable before running code.

Class Lab Part B: Analyze a Demand-Response Game

1. Build a 2-by-2 payoff matrix for two households.
2. For each cell, ask whether A wants to switch.
3. For each cell, ask whether B wants to switch.
4. Mark all pure-strategy Nash equilibria.
5. Discuss whether the equilibrium is fair, reliable, or Pareto efficient.

Research taste

In real grids, human behavior is part of the system model.

AI Collaboration: Good Prompts vs. Bad Prompts

Weak prompt

“Write my regression and Nash equilibrium homework.”

This skips the reasoning and hides the definitions.

Strong prompt

“Here is my MSE formula. Can you help me check whether I am subtracting prediction from data consistently?”

This uses AI as a checker, not a replacement.

Exit Ticket

Answer these before leaving:

1. What does it mean to fit a predefined parametric function?
2. Why can linear regression be written as an optimization problem?
3. What is Taylor expansion trying to approximate?
4. What is a Nash equilibrium in one sentence?
5. Why can Nash equilibrium differ from a socially best outcome?

Homework for This Week

Manual programming training

Complete a small Python script with masked TODO sections for MSE fitting and pure-strategy Nash equilibrium detection.

Main requirements

- Implement MSE for a fitted line without using `sklearn.metrics`.
- Search several slope/intercept candidates and choose the lowest loss.
- Compute a first-order Taylor approximation at one point.
- Detect pure-strategy Nash equilibria in a 2-by-2 payoff matrix.

Feynman Technique

Write a beginner-friendly note explaining fitting-as-optimization or Nash equilibrium using a smart-grid example.

Key Takeaways

1. Fitting means choosing parameters so a predefined function matches data.
2. Linear regression minimizes prediction error, often using mean squared error.
3. Taylor expansion is a local approximation that helps us reason about complicated functions.
4. Neural-network training is also parameter optimization, just at a much larger scale.
5. Game theory studies strategic choices where each player's result depends on other players.
6. Pareto thinking helps us discuss tradeoffs among cost, comfort, reliability, and fairness.